

WARNING LETTER

Sun Pharmaceutical Industries Limited

MARCS-CMS 677337 — JUNE 18, 2024

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VIA EMAIL WITH READ RECEIPT

Product:

Drugs

Recipient:

Mr. Dilip Shanghvi
Managing Director
Sun Pharmaceutical Industries Limited
Sun House, Plot No. 201 B/1
West Express Highway Goregaon East
Mumbai 400063 Maharashtra
India

[✉ Dilip.Shanghvi@sunpharma.com \(mailto:Dilip.Shanghvi@sunpharma.com\)](mailto:Dilip.Shanghvi@sunpharma.com)**Issuing Office:**

Center for Drug Evaluation and Research | CDER
United States

Feedback

Warning Letter 320-24-48

June 18, 2024

Dear Mr. Shanghvi:

The U.S. Food and Drug Administration (FDA) inspected your drug manufacturing facility, Sun Pharmaceutical Industries Limited, FEI 3004561553, at Survey No. 1012, Dadra-396 193, U.T. of Dadra and Nagar Haveli and Daman and Diu, from December 4 to 15, 2023.

This warning letter summarizes significant violations of Current Good Manufacturing Practice (CGMP) regulations for finished pharmaceuticals. See Title 21 Code of Federal Regulations (CFR), parts 210 and 211 (21 CFR parts 210 and 211).

Because your methods, facilities, or controls for manufacturing, processing, packing, or holding do not conform to CGMP, your drug products are adulterated within the meaning of section 501(a)(2)(B) of the Federal Food, Drug, and Cosmetic Act (FD&C Act), 21 U.S.C. 351(a)(2)(B).

We reviewed your January 9, 2024 response to our Form FDA 483 in detail and acknowledge receipt of your subsequent correspondence.

During our inspection, our investigator observed specific violations including, but not limited to, the following.

1. Your firm failed to clean, maintain, and, as appropriate for the nature of the drug, sanitize and/or sterilize equipment and utensils at appropriate intervals to prevent malfunctions or contamination that would alter the safety, identity, strength, quality, or purity of the drug product beyond the official or other established requirements (21 CFR 211.67(a)).

You failed to adequately clean and maintain your equipment used for drug product manufacturing. For example, our investigator observed approximately 450 mL of a (b)(4) stagnant liquid within the (b)(4) of your nondedicated (b)(4). Your chemical analysis of the unknown liquid using liquid chromatography mass spectrometry (LC/MS) showed numerous peaks. In addition, the microbiological analysis identified too numerous to count (TNTC) colony forming units for total aerobic microbial count (TAMC) and total yeast and mold count (TYMC) testing. You used this nondedicated (b)(4) to manufacture over (b)(4) different drug products.

In your response, you state that a faulty (b)(4) valve was the root cause for the observed (b)(4) liquid accumulating in the (b)(4). The (b)(4) valve failed to completely close and seal when you last disassembled it on June 26, 2023. You acknowledge you discovered this equipment malfunction after initiating an investigation as a result of our investigator's findings. You state that previously manufactured drug products, including (b)(4) active pharmaceutical ingredient (API) and its degradants, as well as (b)(4) API, were identified in the liquid. You also identified microorganisms *Sphingomonas paucimobilis*, *Bordetella bronchiseptica*, and *Pseudomonas stutzeri* in the liquid.

We acknowledge you recalled (b)(4) batches and rejected (b)(4) batches of drug products manufactured on the (b)(4) between June 2023 and December 2023.

However, your response does not sufficiently address the cross-contamination issue. You failed to provide a full characterization of each unknown peak from your chemical analysis. It remains imperative that your investigations are comprehensive to prevent recurrence of inadequate equipment maintenance. We also understand from your recent correspondence that a third-party consultant identified numerous gaps in your equipment maintenance program, but you conclude no critical or major findings were noted.

It is your responsibility to ensure your equipment maintenance program is comprehensive and includes appropriate assessment of equipment failures and their impact to product quality.

In response to this letter, provide:

- A comprehensive, independent retrospective assessment of your cleaning effectiveness to evaluate the scope of cross-contamination hazards. Include other manufacturing equipment that may have been improperly cleaned, evaluate the identity of any residues, and assess whether cross-contaminated products may have been released for distribution. The assessment should identify any inadequacies of cleaning procedures and practices, and encompass each piece of manufacturing equipment used to manufacture more than one product.
- A corrective action and preventive action (CAPA) plan, based on the retrospective assessment of your cleaning program, that includes appropriate remediations to your cleaning processes and practices, and timelines for completion. Provide a detailed summary of vulnerabilities in your process for lifecycle management of equipment cleaning. Describe improvements to your cleaning program, including enhancements to cleaning effectiveness; improved ongoing verification of proper cleaning execution for all products and equipment; and all other needed remediations.
- Your CAPA plan to implement routine, vigilant operations management oversight of facilities and equipment. This plan should ensure, among other things, prompt detection of equipment/facilities performance issues, effective execution of repairs, adherence to appropriate preventive maintenance schedules, timely technological upgrades to the equipment/facility infrastructure, and improved systems for ongoing management review. Your plan should also ensure that appropriate actions are taken throughout the company network.

- Your CAPA plan to comprehensively address any gaps identified by your third-party consultant from the assessment of your equipment maintenance program including, but not limited to, inadequate quality unit (QU) oversight, inadequate trending of preventive maintenance, and lack of detailed procedures.
- Your specific activities that are being undertaken to ensure effective remediation of the conditions that caused the specific cross-contamination incidents discussed above. Provide an independent review by your consultant that determines the effectiveness of your CAPA, including but not limited to:
 - o Enhancements to cleaning and maintenance procedures including determination of specific frequencies and locations to be cleaned in all relevant equipment (e.g., (b)(4))
 - o Adequacy of analysis of equipment maintenance and repair history for all (b)(4) equipment
 - o Sufficiency of the evaluation of all significant manufacturing equipment including, but not limited to, the (b)(4) equipment, to identify other sources of contamination
 - o Whether improvements to your analytical methodology to identify residual carryover were adequate
 - o Whether adequate investigations into other unknown (unidentified) peaks detected in the unknown liquid in the (b)(4) were performed
 - o Assessment whether the scope of the investigation and its related CAPA were sufficient

2. Your firm failed to thoroughly investigate any unexplained discrepancy or failure of a batch or any of its components to meet any of its specifications, whether or not the batch has already been distributed (21 CFR 211.192).

Your investigations of out-of-specification (OOS) results were inadequate because they lacked scientific rationale for root cause determinations. For example, you initiated an OOS investigation for an unknown impurity detected during related substances testing of a 12-month stability sample of (b)(4) and (b)(4) tablets. The specification for any unknown impurity for (b)(4) is not more than (b)(4)%, however, your initial test yielded a result of (b)(4)%. Your investigation concluded, without adequate scientific justification, that dirty glassware was the probable root cause. Upon discussion with our investigator, you reopened your investigation and concluded the root cause to be contamination from another product, (b)(4), that was also tested during the same time period. However, this conclusion was not scientifically justified because the retention time of (b)(4) did not match that of the unknown impurity peak.

In your response, you state you performed additional testing of more drug products tested in the same laboratory during the same time period when the OOS result occurred. You identified the root cause as laboratory contamination and the unknown impurity to be (b)(4). In addition, you state you have hired a third-party consultant to review invalidated OOS results generated between April 1, 2019, and December 26, 2023, to determine if the decisions made to invalidate those results were scientifically justified.

Your response is inadequate. You did not explain why root cause determinations in the OOS investigation were accepted by your firm multiple times, although insufficiently justified.

Your quality system failed to ensure appropriate investigations to determine root causes, evaluate the impact of OOS results, and ensure effective CAPA. We recommend that the retrospective review also includes an evaluation of the data associated with pending and approved drug application submissions.

Your OOS investigations conclusions fail to include sufficient rigor and scope in determining root causes and the extent of deviations and drug product impact.

For more information about handling failing, out-of-specification, out-of-trend, or other unexpected results and documentation of your investigations, see FDA's current guidance document *Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production* at <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/investigating-out-specification-oos-test-results-pharmaceutical-production-level-2-revision>.

In response to this letter, provide:

- A retrospective, independent review of all invalidated OOS (including in-process and release/stability testing) results for U.S. products for the last three years from the initial date of inspection and a report summarizing the findings of

the analysis, including the following for each OOS:

- o Determine whether the scientific justification and evidence relating to the invalidated OOS result conclusively or inconclusively demonstrates causative laboratory error.
 - o For investigations that conclusively establish laboratory root cause, provide rationale and ensure that all other laboratory methods vulnerable to the same or similar root cause are identified for remediation.
 - o For all OOS results found by the retrospective review to have an inconclusive or no root cause identified in the laboratory, include a thorough review of production (e.g., batch manufacturing records, adequacy of the manufacturing steps, suitability of equipment/facilities, variability of raw materials, process capability, deviation history, complaint history, batch failure history). Provide a summary of potential manufacturing root causes for each investigation, and any manufacturing operation improvements.
- A comprehensive review and remediation plan for your OOS result investigation systems. The CAPA should include but not be limited to addressing the following:
 - o QU oversight of laboratory investigations
 - o Identification of adverse laboratory control trends
 - o Resolution of causes of laboratory variation
 - o Initiation of thorough investigations of potential manufacturing causes whenever a laboratory cause cannot be conclusively identified
 - o Adequately scoping of each investigation and its CAPA
 - o Revised OOS investigation procedures with these and other remediations
 - A comprehensive, independent assessment of your overall system for investigating deviations, discrepancies, complaints, OOS results, and failures. Provide a detailed action plan to remediate this system. Your action plan should include, but not be limited to, significant improvements in investigation competencies, scope determination, root cause evaluation, CAPA effectiveness, QU oversight, and written procedures. Address how your firm will ensure all phases of investigations are appropriately conducted.
 - An independent assessment and remediation plan for your CAPA program. Provide a report that evaluates whether the program includes effective root cause analysis, ensures CAPA effectiveness, analyzes investigations trends, improves the CAPA program when needed, implements final QU decisions, and is fully supported by executive management.
 - A comprehensive, independent assessment of your laboratory practices, procedures, methods, equipment, documentation, and analyst competencies. Based on this review, provide a detailed plan to remediate and evaluate the effectiveness of your laboratory system.

Repeat Observation at Facility

On March 29, 2019, your firm was cited for inadequate cleaning and maintenance of manufacturing equipment. On April 13, 2017, your firm was cited for inadequate investigations of laboratory events regarding the recurrence of extraneous unidentified peaks. As a result of your 2017 inspection, you were issued a Response Letter dated September 29, 2017, that stated contaminated glassware should not be a frequent justification for rejecting or discarding aberrant result. Furthermore, the letter stated you should assess and identify additional CAPA to ensure clean laboratory glassware is utilized.

You described specific remediations for these observations in your responses.

Repeat Violations at Multiple Sites

FDA also cited a similar CGMP violation at other facilities in your company's network. Sun Pharmaceutical Industries Ltd., (Halol site), FEI 3002809586, was issued a warning letter dated December 15, 2022 (later amended and dated October 16, 2023), for violation of, among other items, 21 CFR 211.67(a) for inadequate cleaning and maintenance of a **(b)(4)**. You committed to conduct a comprehensive assessment of your global manufacturing operations across your company's network to ensure conformance to FDA requirements.

These repeated failures at multiple sites demonstrate that management oversight and control over the manufacture of drugs is inadequate.

We note the initial steps that you outlined in your written response to this inspection, including changes in key senior leaders at corporate and site levels, and modifications to your corporate structure that are intended to enhance quality oversight.

Your executive management remains responsible for fully resolving all deficiencies and ensuring ongoing CGMP compliance.

Ineffective Quality Systems

Significant findings in this letter demonstrate that your firm does not operate an effective quality system in accord with CGMP. In addition to the lack of effective management oversight of your production and laboratory operations, we found your QU is not enabled to exercise proper authority and/or has insufficiently implemented its responsibilities. Executive management should immediately and comprehensively assess your company's global manufacturing operations to ensure that your systems, processes, and products conform to FDA requirements.

CGMP Consultant Recommendation

We strongly recommend that your firm engage a consultant qualified as set forth in 21 CFR 211.34 to assist your firm in meeting CGMP requirements.

Your use of a consultant does not relieve your firm's obligation to comply with CGMP. Your firm's executive management remains responsible for resolving all deficiencies and systemic flaws to ensure ongoing CGMP compliance.

Conclusion

The violations cited in this letter are not intended to be an all-inclusive list of violations that exist at your facility. You are responsible for investigating and determining the causes of any violations and for preventing their recurrence or the occurrence of other violations.

If you are considering an action that is likely to lead to a disruption in the supply of drugs produced at your facility, FDA requests that you contact CDER's Drug Shortages Staff immediately, at drugshortages@fda.hhs.gov, so that FDA can work with you on the most effective way to bring your operations into compliance with the law. Contacting the Drug Shortages Staff also allows you to meet any obligations you may have to report discontinuances or interruptions in your drug manufacture under 21 U.S.C. 356C(b). This also allows FDA to consider, as soon as possible, what actions, if any, may be needed to avoid shortages and protect the health of patients who depend on your products.

Correct any violations promptly. FDA may withhold approval of new applications or supplements listing your firm as a drug manufacturer until any violations are completely addressed and we confirm your compliance with CGMP. We may re-inspect to verify that you have completed corrective actions to any violations.

Failure to address any violations may also result in the FDA refusing admission of articles manufactured at Sun Pharmaceutical Industries Ltd., Dadra, into the United States under section 801(a)(3) of the FD&C Act, 21 U.S.C. 381(a)(3). Articles under this authority that appear to be adulterated may be detained or refused admission, in that the methods and controls used in their manufacture do not appear to conform to CGMP within the meaning of section 501(a)(2)(B) of the FD&C Act, 21 U.S.C. 351(a)(2)(B).

This letter notifies you of our findings and provides you an opportunity to address the above deficiencies. After you receive this letter, respond to this office in writing within 15 working days. Specify what you have done to address any violations and to prevent their recurrence. In response to this letter, you may provide additional information for our consideration as we continue to assess your activities and practices. If you cannot complete corrective actions within 15 working days, state your reasons for delay and your schedule for completion.

Send your electronic reply to CDER-OC-OMQ-Communications@fda.hhs.gov. Identify your response with FEI 3004561553 and ATTN: Christina Alemu-Cruickshank.

Sincerely,
/S/

Francis Godwin
Director
Office of Manufacturing Quality
Office of Compliance
Center for Drug Evaluation and Research

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